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MILITARY SPECIFICATION

OPERATING AND TREATMENT UNIT, DENTAL, FIELD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable, air-operated, dental operating unit with carrying case, designed to provide field dental treatment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
L-P-390	-	Plastic, Molding and Extrusion Material, Polyethylene and Copolymers.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-P-116 - Preservation, Methods of.
MIL-M-7298 - Manual, Technical, Commercial Equipment.
MIL-L-10547 - Liners, Case, and Sheet; Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.
MIL-STD-810 - Environmental Test Methods.

(Copies of specifications, and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS are the issue of the nongovernment documents which are current on the date of the solicitation.

U.S. DEPARTMENT OF COMMERCE
NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein. When materials are not specifically designated, those used shall be of high commercial quality and entirely suitable for the purpose intended.

3.2 Description. The unit shall be a portable, air operated, dental operating and treatment unit, capable of operation as a mobile or stationary unit. The unit shall include a handpiece control system for two handpieces, an oral evacuation system, a self-contained water system, a three-way syringe system, a control cabinet, a collapsible support frame, an air line, an accessory kit, service manuals, a carrying-storage case and have the capability of operating when attached to an external source of air, water and waste removal.

3.3 Handpiece control system.

3.3.1 Air coolant control. The handpiece air coolant flow shall be controlled by a flow control valve located on the front panel. The valve shall be capable of regulating the air flow from complete shutoff to full flow. The foot control valve shall turn the air flow on and off.

3.3.2 Water coolant control. The handpiece water coolant flow shall be controlled by an on-off valve and two variable flow control valves (one for each handpiece) mounted on the front panel. The water coolant control system shall be designed to shut off the water flow when the straight slow speed dry handpiece is activated. The foot control valve shall turn the water flow on and off.

3.3.3 Air pressure gauge. A pressure gauge that indicates the handpiece drive air pressure shall be mounted on the front panel. The pressure gauge shall be graduated from 0 to 100 psig in 1 or 2 psi increments.

3.3.4 Control block. Each handpiece shall have an individual control block or a control manifold sequentially mounted to a main control block. The control system shall be capable of sustained operation using water and air having particulates smaller than 5 microns. The control system shall shut off flow to one handpiece when the other handpiece is activated.

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3.3.5 Foot control valve. The foot valve shall regulate the handpiece drive air flow and the handpiece speed. The valve shall actuate the air and water coolants when the handpiece drive air pressure reaches a maximum pressure of 9 psig. A chip blower shall provide a jet of air when activated. The foot control shall have a cast metal base with anti-skid pads and a scratch resistant cover. The foot controller hose shall have a length of 5'-3" ± 3 inches. The foot control valve shall be capable of varying the handpiece speed by 5 psi increments as indicated on the handpiece pressure gauge when tested in accordance with 4.5.4.1.

3.3.6 Handpiece flush. A handpiece flush system shall be provided. A momentary flush valve, activated by the handpiece control, shall supply water through the handpiece water tube for flushing the handpiece. The handpiece drive and coolant shall not be activated when the flush valve is opened.

3.4 Oral evacuation system. The oral evacuation system shall consist of a saliva ejector, high velocity evacuator and a muffler-collector.

3.4.1 Saliva ejector. The saliva ejector shall operate under a vacuum using a venturi. The venturi shall be replaceable in the field and shall fit into a canister. The canister shall have a 24 mesh filter screen fabricated from 0.0014 inch brass wire.

The hose and flexible tip shall accept and operate with a plastic saliva mouthpiece, (see 6.4.a). The saliva ejector control located on the front panel shall regulate the amount of suction generated. The maximum air consumption shall be 1.5 cubic feet per minute at 80 ± 1 psig when tested in accordance with 4.5.4.3. The saliva ejector hose shall be flexible and 5'-3" ± 3 inches long.

3.4.2 High velocity evacuator. The evacuator shall operate under a vacuum using a venturi and shall provide a minimum of 6.6 ft ^{3}/min of air at the handpiece tip using a maximum of 2.5 ft ^{3}/min of air at a pressure of 80 ± 1 psig when tested in accordance with 4.5.4.2. The handpiece shall have a push button designed to turn the vacuum on or off. The vacuum shall be adjustable both at the front panel and at the handpiece. A flexible semi-rigid spiral wound hose 5'-3" ± 3 inches in length shall connect the evacuator to the waste system. The handpiece shall be provided with three 7/16 inch O.D. by 6 1/2 inch long stainless evacuator tips. A dry stainless steel cuspidor, designed to be attached to the evacuator handpiece, shall be provided with the unit.

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3.4.3 Muffler-collector. The muffler-collector shall separate and collect the waste while allowing the air to be vented to the atmosphere. The muffler-collector shall be designed to lower the noise level generated by the air.

The muffler-collector shall have connections to the high velocity evacuator tube, the vent hose and the saliva ejector hose. All openings and tube connectors shall be capable of being sealed for transport or storage. The vent hose shall be protected by a 0.030 inch plastic screen on 0.065 inch centers. A 64 ounce removable rigid plastic waste collector bottle shall be attached to the collector.

3.5 Self-contained water system.

3.5.1 Water system controls. An on-off switch shall facilitate pressurization of the water tank using 80 psig air. The water system shall include a water regulator designed to control the water pressure at 40 ± 5 psig. The regulator shall have a water filter. The water system shall have a check valve to prevent water from back flowing into the air system. The front panel shall have a standard quarter inch female quick disconnect outlet for attachment to an auxiliary water system.

3.5.2 Handpiece coolant water control. The unit shall not aspirate fluids back into the tubing when tested as specified in 4.5.4.5.

3.5.3 Water tank. The water tank shall be fabricated from austenitic stainless steel and have a 64 ounce capacity. The pressurized water tank shall supply the water for the syringe and the handpiece. A pressure vent valve shall be provided on the water tank.

3.6 Three way syringe. The three way syringe shall consist of a tip, syringe head and handle. The syringe tip shall be fabricated from stainless steel. The handle and head shall be fabricated from nickel plated brass or corrosion-resistant steel. The syringe shall produce a fine stream of air, water, or a mist of air and water. The syringe shall provide the means for individual flow control. The syringe tip shall be easily removable. The syringe tip head and handle shall be autoclavable and show no evidence of corrosion when tested in accordance with 4.5.4.4. The tip shall be rotatable for directional adjustment without loosening the attachment to the syringe head.

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The syringe tip design shall incorporate an O-ring for air-water separation and to prevent air leakage. The air and water push button valves marked "A" and "W" shall be easily removable for valve maintenance. The tip shall be 3 1/4 inches ± 1/4 inch long with the distal end bend at an angle of 35° ± 3° 1 inch ± 1/4 inch from the distal end. The syringe head shall be removable from the unit without the use of tools and without removing the tubing. The syringe shall include a hose having a fully extended length of 6'-3" ± 3 inches.

3.7 Instrument holders. The holders shall be non-metallic and designed to securely hold the handpieces, the three-way syringe, saliva ejector and high velocity evacuator handpiece. The handpiece shall be automatically activated when the handpiece is removed from the holder. The holders shall be equipped with a manual lock-out device. Three holders shall be mounted on a common bracket. Three brackets shall be supplied with each unit. Two brackets shall be furnished with three standard holders. The third bracket shall be supplied with two automatic holders and one standard instrument holder. Each bracket shall be removable from its outside mounting and capable of being mounted on the inside of the cabinet for storage. The instrument handles shall be in a vertical position when mounted in the instrument holders. The cabinet body shall have provisions for mounting of the brackets on both the left and right sides of the cabinet near the front panel.

3.8 Control cabinet. The control cabinet shall be fabricated from anodized aluminum with welded corner seams and shall be rectangular. A removable cover shall be provided which will fit snugly on the top of the cabinet body. The cover shall be fabricated from stainless steel or anodized aluminum. The top of the cover shall have a depression designed to hold a removable stainless steel tray, (see 6.4b and 6.4c). The handpieces, syringe, tubing and holders shall be storable together inside the cabinet. The foot controller shall be storable underneath the cabinet body. The bottom of the cabinet body shall provide a mounting place for the collapsible frame. The faceplate on the front of the cabinet shall have permanent identification markings for the controls. The faceplate shall have two handles, one at each end.

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3.9 Collapsible frame.

3.9.1 Upright. The upright shall support and connect the cabinet to the frame base. The upright shall have a sturdy telescoping outer stainless steel tube and an inner anodized aluminum alloy sliding member. The upright shall be removable without the use of tools for transport or storage in the case. The height of the cabinet, measured from the top of the cabinet to the floor, shall be adjustable between 23 1/2 inches ± 1/2 inch and 32 1/2 inches ± 1/2 inch. The cabinet shall rotate on the upright and shall also have provisions for locking in place.

3.9.2 Base. The base shall be of sturdy "H" frame design and designed to prevent tipping of the unit during use when tested in accordance with 4.5.4.6. The base shall be constructed from anodized aluminum alloy tubing and shall be of welded or bolted construction. Each leg of the base shall be designed to accommodate 2 height adjustable rest pads furnished with the unit for stationary operation and two ball casters furnished with the unit for mobile use. Each end of the legs shall have a cushioning bumper.

3.10 Case. The unit shall be furnished partially or fully disassembled in a rigid, rotationally molded lightweight, vaporproof polyethylene carrying-storage case. The polyethylene shall meet the requirements of L-P-390, for type I, Class M, Grade 2, polyethylene. The case shall be equipped with an air relief valve. The inside of the case shall be designed and cushioned to prevent movement or damage of the dissembled unit when the sealed case is handled. The case, with the unit packed inside, shall be a nominal 20-1/2 inches in length, 17-1/2 inches in width, and 11 inches in height and shall weigh a maximum of 72 pounds.

3.11 Utility hose. The unit shall be supplied with a 10 foot, 3/8 inch O.D. uncoiled braided reinforced air supply hose with Imperial-Eastern 3/8 inch quick disconnects, male on one end and female on the other end. The unit shall have a quick disconnect connection for each of the utilities - air, water and vacuum. These connections shall be designed to hook up to outside sources of utilities.

3.12 Handpieces for rotary instruments. One Star Dental Titan-2 plus motor #3055 with Titan-2 straight nose cone attachment #57506 low speed handpiece, (see 6.4.d) and one Star Futura-2 #53874 (with chip blower) high speed handpiece, (see 6.4.e) shall be furnished with the unit. The handpiece hoses shall be coiled and have a fully extended length of 6 feet 3 inches ± 3 inches. The hoses shall be four hole and shall be designed for the attachment to the handpieces referenced (see 6.4.d and 6.4.e).

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3.13 Accessory kit. The unit shall be provided with an accessory kit containing the necessary repair parts and tools for routine maintenance and repair of the unit. The kit shall, as a minimum, include the following components: a three way syringe service kit, 2 syringe tips, air filter-regulator service kit, control block service kit, plastic sleeve tool, air vacuum system button assembly service kit, male 1/4 inch quick-disconnect fitting.

3.14 Environmental testing. The unit shall, when tested in accordance with 4.5.4.7, meet the requirements of MIL-STD-810 for the following: High Temperature Method 501 Procedure I, Low Temperature Method 502 Procedure I, Vibration Method 514 Category 3 - Loose Cargo Transport and Shock Method 516 Procedure II. The vibration test period shall be 3 hours. A certificate stating that the units were tested and found in compliance with the requirements of MIL-STD-810 shall be submitted to the contracting officer.

3.15 Operation. The dental unit shall operate without malfunctioning or any reduction in performance when tested in accordance with 4.5.4.8.

3.16 Stability. The unit shall not tip over when tested in accordance with 4.5.4.6.

3.17 Finish. All surfaces shall be clean and free of foreign matter, flux, or welding residue. The finish of components shall be as specified, the finish shall be entirely suitable for the purpose intended. Finish on edges and surfaces shall be uniform and free from burrs, slivers, sharp edges, cracks, or other defects. All exposed surfaces shall be able to withstand repeated exposure to disinfectants without deterioration of function or esthetics. No trim shall be attached such that its partial or complete loss will interfere with disinfection.

3.17.1 Plating and coatings. Shall be uniform in thickness and appearance; shall be free from blisters, peeling, visible porosity, or other defects. The base metal to be plated shall be free from flaws or defects which will be detrimental to the final finish and shall be subjected to such undercoating, polishing, cleaning, dipping, and plating procedures as are necessary to yield deposits with the desired function, quality, and appearance.

3.18 Workmanship. Workmanship shall be in accordance with best commercial practice. All components shall be clean, well finished, and free from defects which detract from their appearance or impair their serviceability.

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3.18.1 Castings. Castings shall be sound and free from repairs, warping, or other defects that may render the casting unsound for use. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste shall not be permitted.

3.18.2 Welding, brazing and soldering. All welding, brazing, and soldering shall be durable and sound in every detail. Welding shall be in accordance with the latest practices recommended by the American Welding Society. Components shall be joined without any resultant distortion and shall be free from twists or strains.

All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All aluminum parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

Exposed welds shall be smooth and blend with adjacent surfaces. All welds shall be cleaned, discoloration removed, and be free from slag and spatter before finishing. Welds shall be ground, only to the degree necessary to remove sharp edges or projections which may cause defects to adjacent surfaces or can be injurious to the touch.

3.18.3 Machine work. Machining shall be in accordance with best commercial practice. All bolts, machine screws, and rivet holes shall be sized appropriately for the size screw or bolt used. All bolts, nuts, screws, and fasteners shall be completely fastened and tight.

3.18.4 Identification marking. Each case shall be marked with the following information:

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DENTAL OPERATING AND TREATMENT UNIT, FIELD
Name or registered trademark of the manufacturer
Contract or Purchase Order Number
Serial Number
Letters "US"

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The marking of the case shall be located on the outside top face of the carrying case. In addition to the above marking, each case (top and bottom portion) shall be marked with the serial number of the case on the mating metal extruded portions of the cases so that the numbers can be matched to identify the mating portions of the case.

3.19 Tolerances. Standard commercial tolerances shall be permissible on material gage sizes. Tolerances shall be held to a minimum where accuracy is of paramount importance for proper operation and interchangeability of component parts. Assemblies shall be precision-fitted for durable, smooth, and consistent functioning in keeping with first class workmanship throughout.

3.20 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.21 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation and maintenance, and lists of component parts. The manual shall also include directions for hygienically cleaning, disinfecting or sterilizing applicable components of the unit. The manual shall comply with the requirements of MIL-M-7298.

As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within 2 years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

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3.22 Disinfection. All exterior parts of the unit shall be capable of being hygienically clean in accordance with the manufacturer's instructions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, or does it commit the Government to acceptance of defective material.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

4.2.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Governments request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.2.2 Certificates of quality. Certificate of quality supplied by the manufacturer of the steel may be furnished in lieu of actual performance of composition testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. The unit of product for examination purposes shall be one field dental operating unit.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product for test purposes shall be one field dental operating unit. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests.

Characteristic	Requirements	Test procedure	Inspection level	AQL (percent defective)
Foot control accuracy	3.3.5	4.5.4.1	S-2	1.0
High velocity evacuator	3.4.2	4.5.4.2	S-2	1.0
Saliva ejector	3.4.1	4.5.4.3	S-2	1.0
Syringe corrosion resistance	3.6	4.5.4.4	S-2	1.0

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TABLE II. Sampling for tests (continued).

Characteristic	Requirements	Test procedure	Inspection level	AQL (percent defective)
Handpiece coolant water control	3.5.2	4.5.4.5	S-2	1.0
Stability	3.16	4.5.4.6	S-2	0.5
Environmental testing	3.14	4.5.4.7	S-2	1.0
Operational test	3.15	4.5.4.8	100%	0.5

4.4 Examination.

4.4.1 Examination of dental unit. The dental unit shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however shall not be restricted to defects listed in table III. Any unit in the sample containing one or more defects shall be rejected if the number of defective units in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Unit design and construction not as specified.
102	Hardware components such as pins, screws and fasteners missing, broken or otherwise damaged.
103	Finish not as specified.
104	Damage or defects on exterior or interior surfaces which effect function or serviceability.
105	Plating missing which effects function.
106	Plating not free from blisters, peeling, visible porosity or other defects.
107	Any component fractured, broken, punctured, torn, bowed, deteriorated, or malformed.
108	Any component misplaced or not in proper alignment.
109	Fastening device requiring loosening or removal is swaged, peened, staked, or otherwise permanently fastened.

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TABLE III. Classification of defects (continued).

Categories	Defects
110	Components missing.
111	Components do not fit or mate properly.
112	Interface fits between components not proper (too loose, too tight/binding).
113	Components not free from defects which may impair their serviceability.
114	Components not free from defects which do not impair their serviceability.
115	Removable components cannot be removed and replaced without difficulty.
116	Components not properly assembled or aligned.
117	Components do not store or remove from case without difficulty.
118	Handpiece lockout device does not work.
119	Service data not provided.
120	Coarse machine, tool or die marks which affect function or serviceability.
Minor	
201	Identification markings not present, not complete, not permanent, not correct.
202	Accessory kit not provided.
203	Operating instructions not provided.
204	Casters do not operate or swivel freely when installed on unit.
205	Total unit weight not as specified.
206	Coarse machine, tool or die marks which do not affect function or serviceability.
207	Surfaces not clean, not free of foreign matter, flux or other defects.
208	Damage or defects on exterior or interior surfaces which do not effect function or serviceability.
209	Plating not uniform in appearance.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.4.1 Foot control accuracy. Operate a handpiece at a medium speed by means of the foot control. Increase and decrease the speed by varying the pressure on the foot valve.

4.5.4.2 High velocity evacuator test. Using air at 80 psig and the assembled unit, the high velocity evacuator shall be tested for air flow rate.

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4.5.4.3 Saliva ejector air consumption. Using air at 80 psig and the assembled unit, the saliva ejector shall be tested for maximum air consumption.

4.5.4.4 Syringe corrosion resistance. The syringe shall show no evidence of corrosion when tested in accordance with the boil test of ASTM F1089.

4.5.4.5 Handpiece coolant water control. The unit shall be tested in accordance with the handpiece coolant water control test in ANSI (American National Standard Institute/ADA (American Dental Association) No. 47, specifications for dental units.

4.5.4.6 Stability. The unit shall be tested for stability in accordance with ANSI/ADA No. 47.

4.5.4.7 Environmental testing. The unit shall be tested in accordance with MIL-STD-810.

4.5.4.8 Operational test. Attach a high speed and a low speed handpiece to the hoses. Provide the necessary utilities and attach the saliva ejector mouthpiece. The handpiece control, oral evacuation, the three-way syringe and self-contained water systems - shall be tested by operating each for 5 minutes then turning it off for 1 minute. This procedure shall be repeated. The unit shall then be disassembled and stored in the carrying case without difficulty.

4.6 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1. Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. One portable dental operating and treatment field unit shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

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5.1.1.3 Unit package. Each unit along with air hose, evacuator tips, cuspidor, accessory kit, and service data shall be positioned and secured within the carrying case in such a manner to prevent damage to the unit and components. The case shall be closed and the closure secured by latching.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for portable dental operating and treatment field unit shall conform to the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Each unit packaged in the carrying case as specified in 5.1.1.3 shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601 overseas type, PPP-B-621, class 2, or PPP-B-636, class weather-resistant. The case shall be cushioned with fiberboard pads or other suitable material to prevent damage to the exterior hardware and to prevent the hardware from puncturing the box. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Each unit packaged in the carrying case as specified in 5.1.1.3 shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-636, class domestic. The case shall be cushioned with fiberboard pads or other suitable material to prevent damage to the exterior hardware and to prevent the hardware from puncturing the box. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. One portable dental operating and treatment field unit shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in length, 52 inches in width, and 54 inches in height. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. One portable dental operating and treatment field unit packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. One portable dental operating and treatment field unit packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.

6. NOTES

6.1 Intended use. This specification covers a portable, air-operated dental operating and treatment unit designed for field use.

6.1.1 Ordering data.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-140-7663	OPERATING AND TREATMENT UNIT, DENTAL FIELD, Portable, Air Operated

6.4 National Stock Number (NSN) reference. This specification references the following items appearing in the Federal Supply Catalog:

a.	6520-00-082-2673	Mouthpiece, Saliva Ejector, Dental, Plastic
b.	6530-00-926-9170	Tray, Instrument, Corrosion-Resisting Steel
c.	6530-00-794-0600	Tray, Medicine, Corrosion-Resisting Steel
d.	6520-00-138-2308	Handpiece, Dental, Air Operated, Straight Nose
e.	6520-00-103-2858	Handpiece, Dental, Air Operated, Contra-Angle, Ultra Speed, Low Torque

6.5 Subject term (key word) listing.

Operating Unit
Treatment Unit
Portable
Air-driven

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2043

Location: NCR Disk #10

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-37945A	DENTAL OPERATING AND TREATMENT UNIT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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